# MILITARY SPECIFICATION

## NETWORK, ANTENNA COMPENSATION CN-1404/ARN-89

This specification is approved for use by EL, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

### 1. SCOPE

1.1 This specification covers the Network, Antenna Compensation CN-1404/ARN-89 which is intended for use with, but not part of, the AN/ARN-89( ) ADF system. The CN-1404/ARN-89 will electronically provide sinusoidal compensation for bearing errors, in 2 degree increments, up to + or - 20 degrees. Maximum compensation will occur at 45 and 225 degrees and minimum will occur at 135 and 315 degrees.

### 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this document to the extent specified herein:

SPECIFICATIONS

Federal

| | |
|---|---|
| NN-P-71 | Pallet, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial) |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to US Army Electronics Command, ATTN: DRSEL-RD-TS-S, Fort Monmouth, N.J. 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5826

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| | |
|---|---|
| QQ-S-781 | Strapping, Steel, Flat and Seals |
| PPP-B-585 | Boxes, Wood, Wirebound |
| PPP-B-601 | Boxes, Wood, Cleated Plywood |
| PPP-B-621 | Boxes, Wood, Nailed and Lock Corner |
| PPP-B-636 | Boxes, Shipping, Fiberboard |
| PPP-B-640 | Boxes, Fiberboard, Corrugated, Triple Wall |
| PPP-C-843 | Cushioning Material, Cellulosic |
| PPP-C-1797 | Cushioning Material, Resilient, Low Density, Unicellular, Polypropylene Foam |
| PPP-F-320 | Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes |
| PPP-S-760 | Strapping, Nonmetallic (and connectors) |
| PPP-T-97 | Tape, Packaging, Industrial, Filament Reinforced |

Military

| | |
|---|---|
| MIL-P-116 | Preservation-Packaging, Methods of |
| MIL-E-5400 | Electronic Equipment, Airborne General Specification for |
| MIL-T-5422 | Testing, Environmental, Aircraft Electronic Equipment |
| MIL-P-11268 | Parts, Materials and Processes Used in Electronic Equipment |
| MIL-M-13231 | Marking of Electronic Items |
| MIL-F-14072 | Finishes for Ground Electronic Equipment |
| MIL-S-19500 | Semiconductor Device, General Specification for |
| MIL-P-28809 | Printed Wiring Assemblies |
| MIL-M-38510 | Microcircuits, General Specification for |
| MIL-P-55110 | Printed Wiring Boards |
| MIL-D-55647 | Direction Finder Set AN/ARN-89( ) |



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STANDARDS

Federal

| | |
|---|---|
| FED-STD-356 | Commercial Packaging of Supplies and Equipment |

Military

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection By Attributes |
| MIL-STD-129 | Marking for Shipment and Storage |
| MIL-STD-147 | Palletized and Containerized Unit Loads, 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base |
| MIL-STD-252 | Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired and Other Devices |
| MIL-STD-454 | Standard General Requirements for Electronic Equipment |
| MIL-STD-461 Notice 4 | Electromagnetic Interference Characteristics, Requirements for Equipment |
| MIL-STD-462 Notice 3 | Electromagnetic Interference Characteristics, Measurements of |
| MIL-STD-781 | Reliability Tests, Exponential Distribution |
| MIL-STD-810 | Environmental Tests Methods |
| MIL-STD-1562 | Lists of Standard Microcircuits |

DRAWINGS

Electronics Command

| | |
|---|---|
| DL-SM-B-614513 | Network, Antenna Compensation CN-1404/ARN-89 |

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)



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## 3. REQUIREMENTS

3.1 Description. The CN-1404/ARN-89 is a self-contained electrical compensation network which provides symmetrical, sinusoidal, compensation. It is used with the AN/ARN-89( ) ADF system and installed between the R-1496( )/ARN-89 and AS-2108( )/ARN-89. The CN-1404/ARN-89 when used with the AN/ARN-89( ) system, permits overall correction in increments of $2^{o}$, and up to $+20^{o}$ correction in the first and third quadrants with a similar minus correction in the second and fourth quadrants.

3.2 Construction. Parts shall be in accordance with MIL-P-11268. The units on order shall be constructed in accordance with the following drawing and data list as specified:

DL-SM-B-614513 Network, Antenna Compensation CN-1404/ARN-89

3.3 First article. The contractor shall furnish first article units of equipment on order as required by the invitation for bids and contract. (See 4.3 and 6.2).

3.4 Finish, protective. (See 4.4) The equipment shall be given protective finish in accordance with Specification MIL-F-14072 and the equipment drawings. This includes finish of hardware, such as handles, hinges, screws, etc., and necessary touch-up after mounting. The final paint film on Type I surfaces shall be lusterless black conforming to Specification MIL-F-14072.



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3.5 Securing of parts.

3.5.1 General. Brackets, lugs, inserts, bolts and other mounting arrangements shall retain items securely when subjected to specified service conditions.

3.6 Marking. (See 4.4)

3.6.1 General. Marking shall conform to Specification MIL-M-13231. Front panel marking shall be in accordance with the equipment drawings.

3.6.2 Visibility. Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.6.3 Serial numbers. The unit(s) listed in paragraph 1.1 shall be serial numbered.



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3.7 Printed wiring assemblies. (See 4.4) Printed wiring assemblies shall be processed for compliance with the requirements specified on the equipment drawings, MIL-P-55110 and MIL-P-28809, in order not to affect life and serviceability of the assemblies.

3.8 Semiconductors (transistors and diodes). (See 4.4) These devices (semiconductors and diodes) shall conform to Specification MIL-S-19500 and the equipment drawings.

3. 8.1 Microcircuits. (See 4.4) Microcircuits shall be selected from MIL-STD-1562 and shall conform to MIL-M-38510.

3.9 Tropicalization of materials. Materials shall be treated as follows:

3.9 .1 Treating materials. Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

3.9 .2 Toxicity. Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation or maintenance of the equipment, or during use of the finished items when used for the purpose intended.

3.9 .3 Flexibility. Treatment shall not affect the flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.



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3.9.4 Statement of treatment. The contractor shall submit for approval, to the contracting officer, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use. (See 6.2)

3.10 Service conditions. (See 4.8) The units shall meet the service conditions as specified in Specification MIL-E-5400 for Class I equipment except as otherwise specified herein, and shall meet the specified requirements herein, after a warm-up period not to exceed 60 seconds.

3.10.1 Temperature-altitude. (See 4.8.1) The unit(s) shall be capable of meeting the temperature-altitude requirements with a non-operating minimum exposure of 24 hours at -65°F, atmospheric pressure, and 4 hours minimum exposure at 160°F and atmospheric pressure. The unit(s) shall be operable and conforming to limits specified from -25°F through +160°F. The unit(s) shall be operable from -65°F to -25°F and no measurement shall degrade by more than 6 dB if specified in dB or by more than a factor of 2 if the limit is specified in anything but dB,the input voltage shall be from 24 to 29 Vdc.

3.10.2 Relative humidity. (See 4.8.2) The unit(s) shall be capable of operating without degradation in specified performance and shall sustain no physical damage during and after prolonged exposures of up to 100 percent humidity.

3.10.3 Sand and dust. (See 4.8.3) The unit(s) shall be capable of withstanding exposure to sand and dust particles as encountered in desert areas.



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3.10.4 Vibration. (See 4.8.4) The unit(s) shall be capable of withstanding vibration as encountered in helicopter operation.

3.10.5 Shock. (See 4.8.5)

3.10.5.1 Bench handling. (See 4.8.5.1) The unit(s) shall be capable of withstanding shock as encountered in bench handling.

3.10.5.2 Equipment. (See 4.8.5.2) The unit(s) shall be capable of withstanding 15g's of shock.

3.10.6 Fungus. (See 4.8.6) The unit(s) shall be capable of withstanding exposure to fungus growth as encountered in tropical climates.

3.10.7 Salt fog. (See 4.8.7) The unit(s) shall be capable of withstanding exposure to a salt-sea atmosphere with no degradation of mechanical and/or physical properties as indicated by flaking, pitting, blistering or otherwise loosened finish or clogging or binding of moving parts.



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3.11 Electrical requirements.

3.11.1 D.C. power requirements. (See 4.10.1) The CN-1404/ARN-89 shall be capable of operating from a 15.0 ±1.0 Vdc which is supplied on the "x" channel and "y" channel signal lines originating from the R-1496( )/ARN-89 receiver. The maximum current draw, when the equipment is operating under standard conditions, shall be no more than 50 ma. The unit shall also be capable of supplying to the loop antenna terminals the same 15.0 ±1.0 Vdc received from the R-1496( )/ARN-89.

3.11.2 "X" channel gain. (See 4.10.2) The gain of the "x" channel shall have a fixed value of 1.20 ±0.03 over the frequency range 0.1 MHz to 3.0 MHz.

3.11.3 "Y" channel gain. (See 4.10.3) The gain of the "y" channel shall be variable in 11 steps as defined in Table 1. In addition, the "y" channel gain shall track the "x" channel gain within ±0.02 over the frequency range of 0.1 MHz to 3.0 MHz. (See Figure 1)

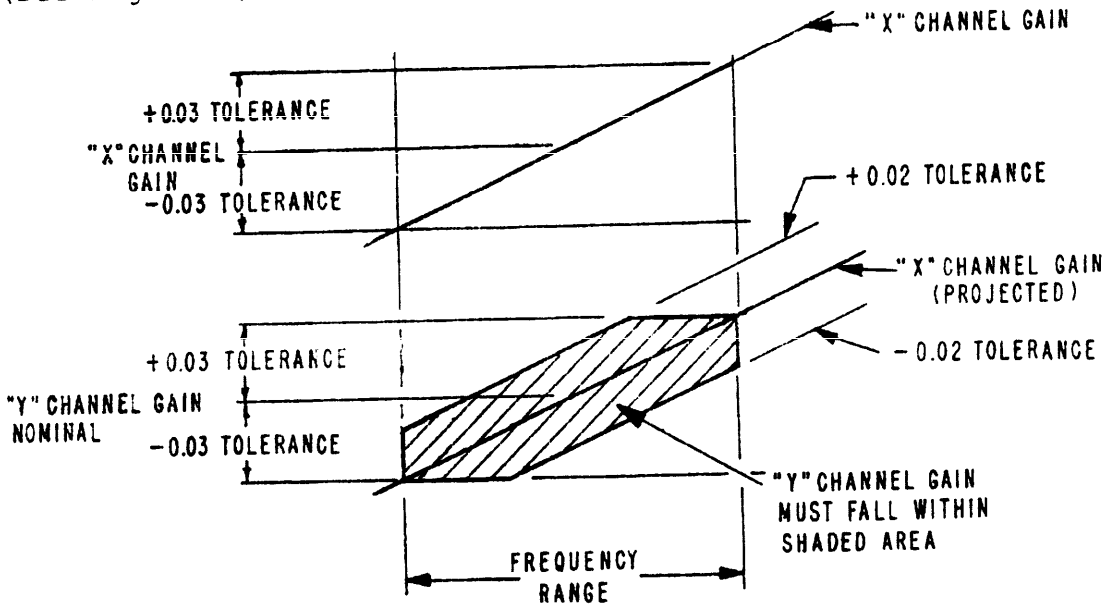



FIGURE 1. Hypothetical example of "Y" channel tracking tolerance



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Table 1

| Switch Position | Gain |
|---|---|
| 0 | 1.200 |
| 1 | 1.132 |
| 2 | 1.064 |
| 3 | 0.996 |
| 4 | 0.928 |
| 5 | 0.860 |
| 6 | 0.792 |
| 7 | 0.724 |
| 8 | 0.656 |
| 9 | 0.588 |
| 10 | 0.520 |

NOTE: Gain tolerance is ±0.03 on all positions.



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3.11.4 Phase shift. (See 4.10.4) The "x" channel to "y" channel phase shift shall not exceed 10 degrees for all settings of gain and over the frequency range 0.1 MHz - 3.0 MHz.

3.12 ADF system requirements.

3.12.1 The following requirements shall apply to the CN-1404/ARN-89 when installed in a standard AN/ARN-89( ) system configuration. Unless otherwise specified the compensation switch of the CN-1404/ARN-89 will be set to zero.

3.12.1.1 System test. (See 4.11.1) The following requirements of MIL-D-55647 shall apply:

| Para. | Requirement |
|---|---|
| 3.15.4.1 | Sensitivity, Comp. Mode |
| 3.15.4.3 | Selectivity, Comp. Mode |
| 3.15.4.4 | Undesired Response, Comp. Mode |
| 3.15.4.8 | Accuracy, Comp. Mode |
| 3.15.4.10 | Noise Level In Headset, Comp. Mode |
| 3.15.4.11 | Audio Control |
| 3.15.4.12 | Harmonic Distortion |
| 3.15.4.13 | Tuning Meter |



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3.12.2 Compensation pattern. (See 4.11.2) The Antenna Compensation Network shall be capable of producing in 2 degree steps, bearing changes from 0 to +20 degrees from an indicated bearing of 45 or 225 degrees and from 0 to -20 degrees at indicated bearings of 135 and 315 degrees. The variation in bearing for each setting shall not exced ±1.5 degrees over the entire frequency range of the receiver.

3.13 Electromagnetic compatibility. The Network, Antenna Compensation CN-1404/ARN-89 shall comply with the requirements of Table V of MIL-STD-461A, Notice 4 as defined hereinafter.

| | |
|---|---|
| RE02 | RS02 |
| RE02.1 | RS03 |
| | RS03.1 |

3.13.1 Bonds, grounds, connectors and cables. Bonding for equipment current path return, RF potentials and shock hazard shall be so installed that expansion, contraction or movement incident to normal service use will not break or loosen the connection. Surface preparation for bonds and grounds shall be accomplished by removing all anodic film, grease, paint, lacquer or other high resistance properties from the immediate area of contact. Bonding impedance shall not exceed 2.5 milliohms. Case or chassis mounted connectors shall have an approved RF conductive finish and shall be installed as above. Bonding impedance of the connector shell to the case shall not exceed 2.5 milliohms. Inter-connecting cables requiring overall shields shall have a minimum of 40 dB shield attenuation to electric fields over the frequency range of 10 kHz to 400 MHz. The shield shall be constructed of tinned copper braid. Connectors used with cables having an overall shield shall have an approved RF conductive finish and provided with EMI backshells for peripheral bonding of the cable shield. RF cables shall be of the double shielded type.



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3.14 Weight. The weight of the CN-1404/ARN-89 shall not exceed 8 oz.

3.15 Burn-in (debugging). (See 4.6) Each unit on order shall be capable of meeting the requirements herein, without subsequent processing after subjection to the burn-in (debugging) as specified.

3.16 Operational inspection. (See 4.15) Each unit on order shall be subjected to the operational inspection as specified, after applicable processes, inspection and tests are completed, to determine that each unit is operable and the Antenna Compensation Network selector switch set to zero prior to packaging.

3.17 Interchangeability. (See 4.14) Like units, assemblies, subassemblies and replaceable parts shall conform to Requirement 7 of MIL-STD-454 and shall be physically and functionally interchangeable, without modification of such items or of the equipment.

3.18 Technical literature and spare parts. Technical literature and spare parts shall be furnished as specified in the contract. Spare parts furnished shall be identical to corresponding parts in the equipment furnished on order.

3.19 Warm-up equipment. (See 4.9) The requirements specified for phase and gain shall be met within 30 seconds after power is applied to the unit(s) at room ambient conditions.



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3.20 Systems safety engineering.

3.20.1 Personnel hazards. Personnel hazards shall be kept to a minimum through compliance of configuration changes and parts selection with Requirement 1 of MIL-STD-454. Compliance with these requirements will be verified through a visual inspection (see 4.16).

3.20.2 Edge rounding exposed. Exposed edges shall be rounded to a minimum radius of 0.04 inch (1 mm), and exposed corners to a minimum of 0.5 inch (13 mm).

3.20.3 Radioactive materials. Radioactive materials shall not be used, (e.g., luminous dials/markings, electron tubes, surge arrestors and lenses).

3.21 Workmanship. The equipment shall be manufactured and assembled in accordance with the equipment drawings (see 3.2). Soldering shall be in accordance with requirement 5 of MIL-STD-454. Workmanhip shall be in accordance with requirement 9 of MIL-STD-454 and the following paragraphs of MIL-P-11268. (See 4.16)

Wiring and cabling including:

Slack
Protection
Splicing
Connections general
Grounding general
Shielding on wire and cable
General requirements for securing of parts
Cleaning prior to welding
Process for welding
Cleaning after welding
Controls
Self tapping screws



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article inspection (does not include packaging. (See 4.3)

b. Inspection covered by subsidiary documents. (See 4.4)

c. Quality conformance inspection:

(1) Quality conformance inspection of equipment before packaging. (See 4.5)

(2) Quality conformance inspection of packaging. (See 4.17)



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4.3 First article inspection. This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for Group A, Group B and Group C (see Tables II and III and IV respectively). The first article inspection will normally be performed in this order: Sample 1: (1) Temperature/altitude; (2) Humidity; (3) Vibration; and (4) Sand and dust: Sample 2: (1) Vibration (2) Temperature/altitude; (3) Humidity; (4) Shock, Bench Handling and Equipment; and (5) Salt fog; Sample 3: (1) Electromagnetic interference; and (2) Explosion proof; Sample 4: (1) Acceleration; and (2) Fungus; other preproduction inspections may precede, follow, or be interspersed between the foregoing.

4.4 Inspection covered by subsidiary documents. The following documents shall be inspected under the applicable subsidiary as part of the inspection before preparation for delivery:

| Item | Where Required |
|---|---|
| Finish | 3.4 |
| Marking | 3.6 |
| Printed Wiring Assemblies | 3.7 |
| Semiconductors (Transistor and Diode) | 3.8 |
| Microcircuits | 3.8.1 |



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4.5 Quality conformance inspection of equipment before packaging. Each unit of each lot of equipment shall be subjected to Burn-in (debugging) (3.15) prior to inspection. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.5. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. Failure to meet any requirement as a result of Burn-in (debugging) shall be classified as a defect and recorded as such. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government, at its discretion, may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 Group A inspection. Each unit of each lot of equipment shall be inspected for conformance to all the inspection and test requirements of Table II. Each lot will be subject to verification, utilizing the procedures of MIL-STD-105, using the general inspection levels and the AQL's indicated in Table II. Group A inspection shall be performed in any order which is satisfactory to the Government, except that the operational inspection shall be last.



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Table II - Group A Inspection

| Inspection | Req. Para. | Insp. Para. | AQL Major | AQL Minor |
|---|---|---|---|---|
| Visual and Mechanical | 3.21 | 4.16 | | |
| Modules: | | | | |
| X Channel Amp | | | .65% | 2.5% |
| Y Channel Amp | | | .65% | 2.5% |
| Chassis, Wired Assy | | | .65% | 2.5% |
| CN-1404/ARN-89 (less cover) | | | 1.0% | 4.0% |
| Electrical: | | | | |
| X Channel Gain | 3.11.2 | 4.10.2 | *1.0% for the Combined Group 1.0% | |
| Y Channel Gain | 3.11.3 | 4.10.3 | | |
| Phase Shift | 3.11.4 | 4.10.4 | | |
| Operational Inspection | 3.16 | 4.15 | | |

*All electrical defects are considered major.



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4.5.2 Group B inspection. This inspection, including sampling, shall conform to Table III and to the procedures for small sample inspection of MIL-STD-105 using the special inspection levels. The AQL shall be 6.5 percent defective and the inspection level shall be S4 for normal, tightened and reduced inspection. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection.

4.5.2.1 Order of inspection within Group B. Group B inspection shall be performed in any order which is satisfactory to the Government.

Table III - Group B Inspection

| Inspection | Req. Para. | Insp. Para. | AQL |
|---|---|---|---|
| Power Requirement, DC | 3.11.1 | 4.10.1 | 6.5% For The Combined Group |
| System Test | 3.12.1.1 | 4.11.1 | |
| Compensation Pattern | 3.12.2 | 4.11.2 | |

4.5.3 Group C inspection. This inspection shall be as listed in Table IV and shall be performed on sample unit(s) that have been subjected to and met Group A and Group B inspections. No defects shall be allowed. Sample unit(s) shall be selected in accordance with 4.5.3.1.



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Table IV - Group C Inspection

| Inspection | Req. Para. | Insp. Para. |
|---|---|---|
| **Subgroup I:** | | |
| Temperature/Altitude | 3.10.1 | 4.8.1 |
| **Subgroup II:** | | |
| Relative humidity | 3.10.2 | 4.8.2 |
| Vibration | 3.10.4 | 4.8.4 |
| Sand and dust | 3.10.3 | 4.8.3 |
| Bench handling | 3.10.5.1 | 4.8.5.1 |
| Equipment | 3.10.5.2 | 4.8.5.2 |
| **Subgroup III:** | | |
| Fungus | 3.10.6 | 4.8.6 |
| Salt Fog | 3.10.7 | 4.8.7 |
| Warm-up equipment | 3.19 | 4.9 |
| Electromagnetic interference | 3.13 | 4.12 |
| Weight | 3.14 | 4.13 |



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4.5.3.1 Sampling for inspection. Unit(s) shall be selected at random for each group of inspections in Table IV as specified in 4.5.3.1.1, through 4.5.3.1.4. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot. Selection of the samples shall be as specified and within 45 calendar days (within 60 calendar days for Subgroup IV) after selection of the samples shall have been subjected to and completed Group C inspections. (See 6.3)

4.5.3.1.1 Subgroup I. For this subgroup, two unit(s) from the first forty (40) production unit(s) shall be selected for the first Group C inspection. For subsequent Group C inspection, two unit(s) from each successive 100 or two per month if less than 100 or fractions thereof are produced monthly shall be selected.

4.5.3.1.2 Subgroup II. For this subgroup, two unit(s) (these unit(s) may be the unit(s) selected for above subgroup 1), from the first forty (40) production unit(s) shall be selected for the first Group C inspection. For subsequent Group C inspection, two unit(s) from each 200 or fraction thereof or two every three months if less than 200 or fraction thereof are produced during three (3) successive months, shall be selected.

4.5.3.1.3 Subgroup III. For this subgroup, two unit(s) from the first forty (40) production unit(s) shall be selected for the first Group C inspection. For subsequent Group C inspection, two unit(s) shall be selected every four (4) months except that fungus and salt fog shall be selected when one-half of the units on order have been accepted by the Government QA representative.



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4.5.4 Non-compliance. The contractor shall immediately report in writing each Group C failure occurrance, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable and (ii) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and Groups A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective.

4.5.5 Reinspection of conforming Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be delivered on the contract or order, provided all damage is repaired, cleaned of all foreign matter, and the sample units are resubjected to and pass Group A inspection.

4.6 Burn-in (debugging). Each unit(s) on order shall be subject to a 48-hour burn-in period at test level E of MIL-STD-781, except that the duration of the vibration shall be at least 50 minutes of each on-time hour. The duty cycle shall be continuous and the temperature cycle shall not be greater than four (4) hours. The rate of temperature change of the thermal medium in both the heating and cooling cycles shall average not less than 5°C/minute. Electrical parameter measurements, as specified in paragraphs 3.11.2 (at 3 frequencies; low, middle and high end of the band), 3.11.3 (at 3 frequencies, low, middle and high end of the band), and 3.11.4 (at 3 frequencies; low, medium and high) will be made in the most efficient order to minimize test time. Measurements shall be taken during the first, one of the middle, and last temperature cycles. Measurements will be taken and recorded at both cold temperature and hot temperature during each measurement period. Measurements shall be taken during periods of vibation and non-vibration. The test records should indicate



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the environment at each reading (e.g., temperature + vibration). Measurements taken at cold temperatures shall start at -65°F and need only be made at one tuned frequency. If measurements taken during a period of no-vibration are not completed within ten minutes, the vibration shall be switched off until the measurements are completed. Each unit shall also be monitored to determine if intermittencies exist. An intermittency of one millisecond or greater duration will be considered to be a failure. Each unit which fails during the burn-in period must be repaired and returned to test until it successfully survives a 48-hour period without failure. Failures which occur during the burn-in will be noted and reported, but will not be counted towards the establishment of the MTBF. Prior to use on contract, the burn-in fixture, test procedures and temperature cycle shall be approved by the procuring activity. The requirements of 3.15 shall be met.

4.7 Standard test conditions CN-1404/ARN-89. Unless otherwise specified inspections shall be performed at room temperature of 77 ±10°F and humidity not to exceed 90%.

a. Phantom fed power supply voltage of 15.0V ± 1.0V on signal output.

b. Interconnecting RF cables: 2 feet.

c. Input/output impedances: 50 Ω.

d. Standard signal inputs.

4.7.1 Standard signal, CN-1404/ARN-89. Unless otherwise specified a standard signal is an unmodulated 10 mv rms (open circuit) signal at each of the following frequencies: 0.1 MHz, 0.5 MHz, 1.0 MHz, 3.0 MHz.

4.7.2 Standard conditions and signals, ADF system. Standard conditions and signals for the ADF system tests of 3.12 shall be those defined in MIL-D-55647. In addition, unless otherwise specified, the compensation switch of the CN-1404/ARN-89 shall be set to zero.



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4.8 Service conditions. The service condition tests for the equipment shall be conducted in accordance with Specification MIL-T-5422 for Class 1 equipment as defined in 3.10. Electrical tests shall apply as specified in Table V. No defects will be allowed.

4.8.1 Temperature-Altitude. Equipment shall be subjected to and meet the operating and non-operating tests of MIL-T-5422 for compliance with 3.10.1.

4.8.2 Relative humidity. The equipment shall be subjected to and meet the humidity test of Specification MIL-T-5422 for compliance with 3.10.2. The unit(s) shall meet the operating test at the completion of each cycle after the third cycle.

4.8.3 Sand and dust. The equipment shall be subjected to the sand and dust test of Specification MIL-STD-810, Method 510.1 for compliance with 3.10.3.

4.8.4 Vibration. The equipment shall be subjected to the vibration test of MIL-T-5422, Procedure I, Curve III of Figure 5.

4.8.5 Shock.

4.8.5.1 Benchhandling. The equipment shall be assembled for servicing and placed in a suitable position for servicing on a solid two-inch fir bench top. The test shall be performed as follows, in a manner simulating shocks liable to occur during servicing:



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a. Tilt up the assembly through an angle of 30 degrees, using one edge of the assembly as an axis, and permit the assembly to drop back freely to the horizontal. Repeat, using other practicable edges of the same horizontal face as axis for a total of four drops.

b. Repeat a. above, with the assembly resting on other faces, until it has been dropped a total of four times on each face on which the assembly could be practically placed for servicing. Tests shall be performed before and after shock, to insure compliance with paragraphs 3.11.2, 3.11.3 and 3.11.4.

4.8.5.2 Equipment. The equipment shall be subjected to the Basic Design and Crash Safety Shock Tests of MIL-T-5422. After these tests the equipment shall comply with the requirements of 3.11.2, 3.11.3 and 3.11.4.

4.8.6 Fungus. The unit(s) including accessories shall be subjected to 28 days of the fungus test specified in Method 508.1 of MIL-STD-810. For gasket or other non-hermetically sealed equipment, the parts and materials of the chassis shall be sprayed with the spore suspension. The interior of the case shall also be sprayed. The equipment shall be subjected to the test in the open condition and as specified for compliance with 3.10.6 and shall not be required to be operated.

4.8.7 Salt fog. The unit(s) including accessories shall be exposed in its normal operating condition to salt fog in accordance with Method 509.1 of MIL-STD-810 for 48 hours. The equipment shall not be opened and shall not be required to be operated. The equipment shall meet the requirement of 3.10.7.

4.9 Warm-up equipment. Set up the unit(s) to check the audio output. Energize the unit(s). Full output shall be noted within 30 seconds. These outputs shall be measured to determine compliance with 3.19.



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TABLE V

| Req. Para. | Test Freq KHz | VIB | | TEMP ALT. | | SHOCK | | HUMID | | SAND & DUST | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Before and After | During | Before and After | During | Before and After | During | Before and After | During | Before and After | During |
| 3.11.2 | 100<br>500<br>1000<br>2000<br>2999 | X | X | X | X | X | N/A | X | X | X | N/A |
| 3.11.3 | 100<br>500<br>1000<br>2000<br>2999 | X | X | X | X | X | N/A | X | X | X | N/A |
| 3.11.4 | 100<br>500<br>1000<br>2000<br>2999 | X | X | X | X | X | N/A | X | X | X | N/A |
| | | Mechanical Visual Inspection & Operation of Controls (See 3.5) | | | | To be conducted before and after each Service Conditions Test | | | | | |



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4.10 Electrical tests.

4.10.1 DC power requirements (See 3.11.1). Terminate both inputs and output with an AC impedance of 50 Ω. Apply a standard signal (see 4.7.1) to the input, and 15.0 Vdc ±1.0 Vdc phanton fed on the output line. With the selector switch in any position, the unit shall not draw more than 50 ma from the 15 V source and will provide a phanton feed 15 Vdc ±1 V thru the signal input lines.

4.10.2 "X" channel gain. Using standard operating conditions and a "X" channel standard signal input measure the "X" channel output at each standard frequency for compliance with 3.11.2.

4.10.3 "Y" channel gain. Using standard operating conditions, measure the "Y" channel output with a 0.5 MHz 10 mv input signal at each switch setting for compliance with Table I. Then measure the output of frequencies of 0.1 MHz, 1 MHz and 3 MHz for compliance with 3.11.3.

4.10.4 Phase shift. Using standard operating conditions and standard signal input on the "X" and "Y" channels, at each switch position; verify that there is zero phase shift between "X" and "Y" inputs, then measure the phase shift at the "X" and "Y" channel outputs. Perform this test at each standard frequency for compliance with 3.11.4.

4.11 ADF system. The CN-1404/ARN-89 shall be installed in an ADF system configuration for the following tests:

4.11.1 Perform the applicable tests in Section 4 of MIL-D-55647 as required by 3.12.1.1 of this specification. The compensation switch on the CN-1404/ARN-89 will be set to zero and the goniometer of the ADF system will have zero compensation.



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4.11.2 Compensation pattern. Using at least three different standard frequencies (as specified in MIL-D-55647) record the goniometer bearing for each CN-1404/ARN-89 compensation switch setting when the input signal originates from the angles 45, 135, 225, 315 for compliance of 3.12.2.

4.12 Electromagnetic compatibility. The equipment shall be tested for compliance with the requirements of 3.13. Test setups and measurement techniques shall be in accordance with MIL-STD-462, Notice 3 as detailed in a contractor prepared Government approved test plan. Interconnecting cables used during test shall be similar in type, shielding and length to those to be used in an operational installation. Tests shall be performed with equipment simulating normal operation.

4.12.1 Bonds, grounds connectors and cables. Prior to the performance of EMI/EMC tests, the bonding impedance requirements of para 3.13 shall be verified. Bonding impedance of the following items shall be measured with a shallcross Meter Model 670A or equal:

a. All connector shells to chassis frame.

b. Power line filter body to ground and chassis frame.

c. Across all surfaces requiring EMI gaskets.

d. All switches, meters, hinge fasteners, transformers, iron core inductors and all mechanical mating surfaces which become exposed during maintenance and repair.

e. Bond and ground connections.



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4.13 Weight. The unit shall be weighed to determine compliance with 3.14.

4.14 Interchangeability. The mechanical dimensions and electrical performance required by the mechanical gage drawings and electrical test fixture drawing listed below shall be gaged or measured to determine conformance to the physical and functional interchangeability requirements of 3.17. When a listed dimension or performance requirement is not within specified or design limits, it shall be considered a major defect. The following dimensions or performance shall be gaged or measured prior to assembly into the units:

a. Mechanical (dimension). The gage points covered by the drawings listed on the applicable gage lists as listed in the contract for this order.

b. Electrical performance (modules). Prior to assembly into the equipment, modules shall be tested to determine compliance with the performance requirements specified in the applicable test procedures and gage lists as listed in the contract for this order.

4.15 Operational inspection. Under normal operating conditions, apply a 10 millivolt, 0.5 MHz, unmodulated signal to both the "X" and "Y" inputs. Set the compensation switch to "0". Each output shall be 12.00 ±0.50 millivolts.



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4.16 Visual and mechanical inspection. The unit(s) shall be examined for defects listed in MIL-STD-252. Prior to assembly into the unit(s) printed circuit boards shall be examined for compliance with MIL-P-55110 and assemblies, modules and sections shall be examined for compliance with the equipment drawings and MIL-STD-252. The requirements of 3.21 shall be met. An inspection shall be performed to verify compliance with those portions of 3.20 (Systems Safety Engineering) which can be determined visually.

4.17 Quality conformance inspection of packaging. Packaging shall be tested in accordance with MIL-P-116 to determine conformance to the requirements of Section 5. The rough handling test will be performed only when specified (see 6.2C).



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## 5. PACKAGING

5.1 Preservation. Preservation shall be level A, B or Commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Network, Antenna Compensation CN-1404/ARN-89 shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Network, Antenna Compensation CN-1404/ ARN-89 shall be dried in accordance with the applicable procedures of MIL-P-116.

5.1.1.3 Preservative application. No preservative required.

5.1.1.4 Unit packing. Unit packing shall be in accordance with the methods/submethods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. Each technical literature shall be unit packed submethod IC-1.

5.1.1.4.2 Network, Antenna Compensation CN-1404/ARN-89. Each network shall be individually unit packed method III as follows: wrap unit with cushioning conforming to PPP-C-843, type II, class B or PPP-C-1797 to protect all projections and absorb the shock of impact in handling or transit. Place the wrapped unit within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Place the technical literature, unit packed as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the box. Close the box in accordance with the appendix of the box specification.



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5.1.2 Level B. Cleaning, drying, preservative application and unit packing shall be as specified in 5.1.1.

5.1.3 Commercial preservation. Preservation shall be in accordance with FED-STD-356.

5.2 Packing. Packing shall be level A, B or Commercial, as specified (see 6.2). Shipping containers for level A and B shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

5.2.1 Level A. A quantity of Network, Antenna Compensation CN-1404/ARN-89, unit packed as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type 1, finish A.



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5.2.1.1 Unitization. Palletization shall be required when: containers specified in 5.2.1 do not require skids; quantities per destination comprise a pallet load; and container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of containers packed as specified in 5.2.1, except that container strapping may be omitted, shall be placed on a pallet, load type 1, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. The load shall be"bonded" to the pallet by strapping conforming to QQ-S-781, type 1, finish A, or shrink film.

5.2.2 Level B. A quantity of Network, Antenna Compensation CN-1404/ARN-89, unit packed as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-640, class 2, style E, or PPP-B-636, type CF, class weather-resistant. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97 and PPP-S-760, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet-mounted on pallets conforming to NN-P-71, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type 1, finish A, or shrink film. When a single unit is shipped to a single destination no packing is required; the unit container shall serve as the shipping container. Reinforcing shall be as specified above.



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5.2.2.1 Unitization. Palletization shall be required when quantities per destination comprise a pallet load. A quantity of Network, Antenna Compensation CN-1404/ARN-89, unit packed as specified in 5.1, shall be placed on a pallet, load type 1, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5c, or V3c. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type 1, finish A, or shrink film.

5.2.3 Commercial packing. Packing shall be in accordance with FED-STD-356.

5.3 Marking.

5.3.1 Military marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with FED-STD-356.



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6. NOTES

6.1 Intended use. The CN-1404/ARN-89 ADF compensation network is intended for use with the AN/ARN-89( ) ADF system. When the AN/ARN-89( ) goniometer is set for zero compensation the CN-1404/ARN-89 will electronically compensate for sinusoidal type errors up to 24° within the frequency range of 100 kHz to 3.0 MHz.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Level A, Level B or Commercial preservation and packing (see Section 5).

c. When rough handling test is required (see 4.17).

d. First article inspection:

(1) Several sample units of each item cited in Section 1 are generally required so that lengthy environmental tests can be completed on specified sample units while complete performance measurements can be made on one sample unit. (See 3.3).

(2) First article pack(s) as follows:

Makeup of pack(s)
Number of each kind of pack to be submitted.
Inspection to be performed thereon.
(Including rough handling).



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f. Marking and shipping of sample.

g. Place of final inspection.

h. Technical literature required.

i. Quantity of spare parts required.

j. The statement of treatment referenced in 3.9.4 should be submitted as specified in the solicitation/contract. This statement should be submitted to the contracting officer.

6.3 Group C inspection. Approval to ship unit(s) on order may be withheld, at the discretion of the Government pending decision from the procuring activity, until compliance with 4.5.3.1 is complete.

6.4 Location of operational inspection. It is desirable that the operational inspection 4.15 be performed at a location that will minimize handling (which might cause damage to the units) after this inspection. It is recommended that the entire lot (including the sample units) be inspected immediately prior to packaging.

6.5 Definition of Level B. Level B packaging requirements detailed herein are designed to furnish protection against corrosion, deterioration, and physical damage during domestic shipments; covered storage for a period of three years or less; and distribution within the continental limits of the United States.

6.6 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.



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6.7 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive qualify assurance provisions established in the specification and the contract. The amount of verification by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

Custodian:
ARMY-EL

Preparing Activity
ARMY-EL
Project Nr. 5826-A139



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# SPECIFICATION ANALYSIS SHEET

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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-N-19116(EL) NETWORK, ANTENNA COMPENSATION CN-1404/ARN-89

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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